



Fig. 29: AN/SPN-46 radar antennae

beacon responds in X-band for interrogation of the tracking radar. Replies of the beacon are continuously plotted on the ACLS computer to derive the aircraft's location with respect to the glideslope.

Instructions to the autopilot are transmitted in real-time through a UHF datalink. Autopilot has a component called approach power compensator (APC), which increases or decreases the engine's power to vary the altitude according to the glideslope requirement. The pilot has warning lights that give indications to abort landing and try for another attempt. Similarly, at any point of time, the pilot can disengage from autopilot by applying pressure on the control stick to take control of the aircraft.

The irony is that pilots do not use this mode! They say that they do not want a computer to land their aircraft. They instead use the instrumented mode and talk-down mode. So this mode only serves as a fallback option during haze and fog.

Mission planning systems

Air operations are planned taking a large number of inputs from various departments and workspaces of the carrier. The following systems help in that planning.

Aviation data management and

control system (ADMACS). This is a tactical, real-time LAN used for an aviation-related information-management system. This system helps to manage the aircraft's launch and recovery operations. This network, extending to aviation-related spaces of the carrier, facilitates strike operation planners with necessary details about aircraft availability and aircraft under maintenance and repair, among others.

Aviation weapons information management systems (AWIMS). This is a data management system to collect, distribute and display information used to manage weapons inventory in a carrier.

Integrated shipboard information system (ISIS). It is a data processing and display system. It integrates all crucial information like tactical, navigational, air operations and meteorological-related data and displays these in a single usable format. To the captain of the ship, entire data regarding the carrier is displayed. Down the hierarchy, only data relevant to a particular function is displayed at various work centres.

In the fifth and concluding part we will discuss command and control operations as well as cruise control operations.

To be concluded next month

DIGITAL CLAMP METERS

36-AUTO

27-AUTO

36-AUTO (AC/DC)

- 3½ Digit 4000 Counts
- Auto Ranging
- Current Upto 600A AC/DC
- Voltage Upto 600V AC/DC
- Resistance Upto 40MΩ
- Jaw Opening Upto 25mm
- Auto Power Off
- Non Contact Voltage
- Temp., Capacitance, Data Hold, Hz/Duty, REL (Zero)

27-AUTO (AC)

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- Auto Ranging
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- Voltage Upto 600V AC/DC
- Resistance Upto 20MΩ
- Jaw Opening Upto 25mm
- Auto Power Off
- Non Contact Voltage
- Data Hold & Max Function

Meco Meters Private Limited

Plot No. EL-50, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2787 3300 (Board), 2787 3311-16 (Sales)
Fax : 0091-22-2787 3330, 2787 3330
E-mail : sales@mecoinst.com
Web : www.mecoinst.com